Design Specification

Smart Ambulance Green Corridor System

1. System Block Diagram

Request push buttons (x4) → debounce (74HC14) → SR request latch → synchronizer → 74HC148 priority encoder → FSM ↔ timing counter → lamp decoder → 12 LEDs. A 1 Hz clock drives the synchronizers, counter, and FSM.

2. Functional Parameters

Parameter	Value
System clock	1 Hz
Normal green	8 s
Yellow	2 s
All-red clearance	1 s
Emergency green	Held until clear signal; 10 s timeout
Priority order	N > E > S > W (fixed, via 74HC148)
Inputs	4 request push buttons (latched), 1 clear push button, reset, clock
Outputs	12 lamp lines (R/Y/G x 4)
Supply	5 V, about 300 mA max

3. FSM States

State	Name	Lamp output	Duration / exit
S0	NS_G	N,S green; E,W red	8 s, then NS_Y (early exit on request)
S1	NS_Y	N,S yellow; E,W red	2 s, then ALL_RED
S2	EW_G	E,W green; N,S red	8 s, then EW_Y (early exit on request)
S3	EW_Y	E,W yellow; N,S red	2 s, then ALL_RED
S4	ALL_RED	All red	1 s, then EMERG if request pending, else next normal state
S5	EMERG	Winning direction green; others red	Until clear or 10 s timeout, then ALL_RED

Note: ALL_RED must remember whether NS or EW comes next. Add one extra flip-flop or split ALL_RED into two states.

4. Operating Behavior

• Normal mode: NS_G, NS_Y, ALL_RED, EW_G, EW_Y, ALL_RED, repeat.

• A request arriving mid-cycle: finish the current yellow, pass through ALL_RED, then enter EMERG.

• In EMERG, the encoder's 2-bit code selects the approach that shows green; all others show red.

• Pressing CLEAR (or the 10 s timeout) ends EMERG; the FSM passes through ALL_RED and resumes the normal cycle.

• Simultaneous requests: the 74HC148 grants the highest priority first; the rest stay latched and are served next.

• The counter resets on every state change; terminal counts are decoded for 1, 2, 8, and 10 s.

5. Switch-Based Ambulance Detection Design

• Each approach has one push button (REQ_N, REQ_E, REQ_S, REQ_W). Pressing it means an ambulance has arrived on that road.

• Each button has a pull-up resistor, an RC debounce filter, and a 74HC14 Schmitt inverter stage.

• The cleaned signal sets a NAND SR latch (active low, which suits the 74HC148 inputs). The latch holds the request after the button is released.

• A single CLEAR push button simulates the ambulance crossing. The encoder's 2-bit code goes through a 74HC139 decoder so CLEAR resets only the latch of the direction being served.

• A D flip-flop on the system clock synchronizes each request before it reaches the FSM.

6. Test Plan

Test	Expected result
Normal cycle	Correct sequence and timing, no conflicting greens
Single button press, each direction	That direction gets green after current yellow and ALL_RED
Simultaneous requests	Highest priority served first, others served after
CLEAR not pressed	Timeout at 10 s, return to normal
Button bounce	One press produces one clean latched request
Reset	FSM returns to NS_G, latches cleared